

Dylan Pieper

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Profile

Data scientist with 5+ years building end-to-end data systems for healthcare and public health. Designs and ships integrated production pipelines, automated reports, interactive dashboards, and LLM applications. Skilled at translating messy, large, and complex data into decisions for clinical, government, and research audiences.

Skills

- **Methods:** Bayesian/frequentist multilevel modeling, LLM application and evaluation, NLP, psychometrics, mixed methods
- **Languages:** R, Python, SQL (MySQL, PostgreSQL, DuckDB), HTML/CSS/JS
- **Applications:** Shiny, Quarto, Tableau, REDCap, Qualtrics
- **Infrastructure:** Git, GitHub, Docker, Azure, Microsoft Foundry, Posit Connect, Linux (RHEL)

Experience

Research Analyst

Nov 2025 - Present

CISR-UME, School of Medicine and Public Health, University of Wisconsin - Madison, WI

- Designed and shipped an **end-to-end LLM pipeline** on **Azure** and **Microsoft Foundry**, transcribing OSCE video recordings with **Whisper**, cleaning transcripts with **NLP**, and generating and quality-checking pedagogically meaningful feedback at scale
- Built a **Quarto** faculty review interface for comparing AI-generated and original feedback, with a secure **JSON**-based offline data collection workflow to handle sensitive OSCE data
- Built and deployed the Medical Education Research Hub, a production **Shiny** app on **Posit Connect**, the Medical Education Office's central directory of research projects (<https://connect.doit.wisc.edu/meded-research-hub/>)
- Automated data reporting with **Quarto** for teaching faculty and administrative staff

Data Scientist

Sep 2021 - Oct 2025

School of Pharmacy, University of Pittsburgh - Pittsburgh, PA

- Owned data strategy across multiple grant-funded projects, scoping requirements with IT departments, researchers, healthcare providers, and public officials through to delivery
- For a state-funded opioid treatment project, designed **REDCap** integrations of electronic medical records, built automated **ETL pipelines** in **R** and **DuckDB**, and applied **Bayesian multilevel modeling** to predict retention risk across a 250+ site network
- For an Arnold Ventures criminal justice project, built **frequentist multilevel models** of racial disparities in pretrial bail and created interactive **D3** Sankey diagrams tracking case outcomes from arraignment through sentencing
- Engineered a **MySQL** database linking clinical, operational, and geographic data
- Built and deployed data collection tools (**REDCap**, **Qualtrics**) and dashboards (**Shiny**, **Tableau**) serving project teams, treatment sites, and government partners
- Mentored junior data staff through recruitment, onboarding, and professional development

Experience

Field Interviewer

Aug 2021 - Nov 2021

NORC, University of Chicago - Chippewa Falls, Wisconsin

- Conducted structured interviews across diverse locations, maintaining data quality through standardized protocols while upholding confidentiality and research ethics standards
- Managed end-to-end logistics including scheduling, travel, and participant communication to ensure successful data collection across assigned areas

Research Specialist

Aug 2020 - Aug 2021

Michele Gelfand's Culture Lab, University of Maryland - College Park, MD

- Published **frequentist multilevel models** of COVID-19 outcomes across 57 countries in **The Lancet Planetary Health**
- Built and deployed an **NLP Shiny** app to detect and compare threatening language across texts (<https://www.michelegelfand.com/threat-dictionary>)
- Ran randomized experiments on public health message framing using **Qualtrics**

Writing Coach

Aug 2017 - May 2020

University of Northern Iowa - Cedar Falls, IA

- Coached students in writing and statistics and supported ELLs and students with learning disabilities
- Trained new tutors and developed training materials

Education

Master of Arts - Social Psychology

2018 - 2020

University of Northern Iowa - Cedar Falls, Iowa

- Completed courses **Statistical Computing I and II** in R
- Worked as a **Research Assistant** in the Psychoneuroendocrinology Lab
- Worked as a **Teaching Assistant** in undergraduate psychology courses (Introduction to Psychology, Biopsychology, and Psychology of Gender)
- Thesis: "Challenging Social Systems Under the Threat of Pollution" (<https://scholarworks.uni.edu/etd/1028/>)

Bachelor of Arts - Psychology

2014 - 2018

University of Northern Iowa - Cedar Falls, Iowa

Publications

- Gelfand, M. J., Jackson, J. C., Pan, X., Nau, D., **Pieper, D.**, Denison, E., Dagher, M., Van Lange, P. A. M., Chiu C., & Wang, M. (2021). The relationship between cultural tightness-looseness and COVID-19 cases and deaths: A global analysis. *The Lancet: Planetary Health*. [https://doi.org/10.1016/S2542-5196\(20\)30301-6](https://doi.org/10.1016/S2542-5196(20)30301-6)
- Gelfand, M. J., Li, R., Stamkou, E., Denison, E., Fernandez, J., Choi, V., Chatmen, J., Jackson, J. C., **Pieper, D.**, & Dimant, E. (2021). Persuading conservatives and liberals to comply with mask-wearing: An intervention tournament. *Journal of Experimental Social Psychology*. <https://doi.org/10.1016%2Fj.jesp.2022.104299>

Publications

- Dimant, E., Clemente, E. G., **Pieper, D.**, & Azevedo, F. (2022). Politicizing mask-wearing: Predicting the success of behavioral interventions among Republicans and Democrats in the U.S. *Scientific Reports*, 12, 7575. <https://doi.org/10.1038/s41598-022-10524-1>
- **Pieper, D.** (2020). Challenging social systems under the threat of pollution: Replication and extension of Eadeh and Chang (2019) [Master's Thesis, University of Northern Iowa]. UNI ScholarWorks. <https://scholarworks.uni.edu/etd/1028>

Presentations

- **Pieper, D.** (2025, September). From messy to meaningful data: LLM-powered classification in R. *posit::conf*. <https://github.com/dylanpieper/posit-25>
- **Pieper, D.** (2025, June). redquack: An R package for memory efficient REDCap-to-DuckDB workflows. *R/Medicine*. <https://r-consortium.org/posts/redquack-an-r-package-for-memory-efficient-redcap-to-duckdb-workflows>
- Katie, W. & **Pieper, D.** (2024, June). The Pennsylvania medications for opioid use disorder (PA MOUD) technical assistance and quality improvement expansion project. *Rural Health Information Hub*. (Online resource). <https://www.ruralhealthinfo.org/project-examples/1139>
- **Pieper, D.**, Gelfand, M. J., & Denison, E., (2021, February). COVID-19: Meta-norms to wear masks and social distance in the U.S. *Society for Personality and Social Psychology*. (Remote poster session). <https://osf.io/y8cug>
- **Pieper, D.**, Hirano, H., & Harton, H. C. (2020, April). The effect of environmental threats and terrorism/violent-crime threats on political orientation and attitudes. *Midwestern Psychological Association Conference*. Chicago, Illinois. (Conference canceled) ... Also (2020) at the *University of Northern Iowa's College of Social and Behavioral Sciences Conference*. Cedar Falls, Iowa. (Conference canceled). <https://osf.io/3n7r8>
- **Pieper, D.** (2020, February). Challenging the system under the threat of pollution: Low system justification mediates a liberal political shift. *Society for Personality and Social Psychology*. New Orleans, LA. (Poster session) ... Also (2020) at the *University of Northern Iowa's College of Social and Behavioral Sciences Conference*. Cedar Falls, Iowa. (Conference canceled). <https://osf.io/v7nez>
- **Pieper, D.** (2019, April). What's in a name? Measuring the potential influence of medical cannabis labels on attitudes toward its use. *Iowa Psychological Association Conference*, Des Moines, Iowa. (Poster session) ... Also (2019, April) at the *University of Northern Iowa's College of Social and Behavioral Sciences (CSBS) Conference*. Cedar Falls, Iowa. Awarded Best Oral Presentation (2nd place, \$250). <https://scholarworks.uni.edu/agss/2019/all/18>
- **Pieper, D.**, DeSoto, C., & Hitlan, R. (2018, April). The relations between personality and source of dietary protein. *Midwestern Psychological Association Conference*. Chicago, Illinois. (Poster session) ... Also (2017, April) at the *University of Northern Iowa's College of Social and Behavioral Sciences Conference*. Cedar Falls, Iowa.

R Packages

- **redquack**: Transfer REDCap data to a database and use in R without exceeding available memory (<https://dylanpieper.github.io/redquack>)